

254 ENGINEERING SERVICES

Foundation Certification Protocol

Document	Value
Title	Foundation Certification Protocol for Existing Foundations
Document number	254-FC-001
Version	1.0
Issue date	September 29, 2026
Prepared by	Aman Dhakal, P.E., Engineer of Record, 254 Engineering Services
Approval	Aman Dhakal, P.E., Engineer of Record
Applies to	All foundation certification engagements performed by 254 Engineering Services
Supersedes	None. First issue of this service line.

1. Purpose

This protocol establishes how 254 Engineering Services performs foundation certifications of existing foundations so that the engineer can issue a sealed certification letter supported by an evidence package collected by a field technician under the engineer's supervision.

The engineer signs and seals the certification letter personally and the recipient relies on that opinion. Every requirement in this protocol exists so that the engineer can reach a determination from the evidence package alone.

2. Scope

This protocol applies to condition assessment and certification of existing foundations on residential and light commercial structures where a client requires a professional opinion on the condition of the foundation for insurance binding or renewal, a property sale, lending, or a comparable purpose recorded at intake.

This protocol does not cover foundation work on ongoing new construction — the firm does not offer that service at this time; manufactured home foundation certifications, which are covered by their own protocol; structural design; or repair specification. Each of those is a separate service line under its own protocol and engagement terms, where offered. The firm does not perform forensic or cause-of-loss investigation, geotechnical investigation, or subsurface soil testing under this or any protocol.

3. Governing requirements

No state program certifies the condition of an existing foundation in Texas. The certification letter is the engineer's professional opinion of observed condition as of the inspection date, based on a visual, non-invasive assessment. It is not a warranty, a guarantee against future movement, or a representation that the foundation will not require future repair.

4. Roles and responsibilities

Only the Engineer of Record determines the outcome of a job and the wording of the letter. The field technician and the coordinator relay the engineer's written determination without interpretation, addition, or estimate of when a letter will issue.

5. Personnel qualifications

Field Technician

- At least two years of documented experience in foundation inspection, residential construction, or geotechnical fieldwork, or equivalent experience the engineer accepts in writing.
- Completion of the engineer's training on this protocol and Appendix B, with a supervised inspection before independent work.
- Demonstrated proficiency with the firm's elevation survey equipment, including benchmark selection and reading procedure.
- Confined-space awareness training where crawlspace access is required.
- Annual refresher training when the adopted code, this protocol, or Appendix B changes.

Engineer of Record

- Active Texas P.E. license. A copy, with expiration date, is held in the firm's compliance file.

6. Intake

- The coordinator opens a job file for each property. The file records the intake answers and customer uploads required by Appendix A.
- The engineer records the purpose of the letter and the recipient exactly as given at intake. The letter is addressed only to that recipient and issued only for that purpose.
- A yes to any flag question in Appendix A routes the job to the engineer before dispatch. The engineer records accept, accept with conditions, or decline.
- A job is not dispatched until intake is complete and every upload required by Appendix A has been received.
- The engineer records any condition of acceptance in the job file, including any item the technician must capture beyond Appendix B.

7. Scheduling and dispatch

- The coordinator schedules an accepted job within two business days. The engineer may set a different interval for a job in the job file.
- The technician confirms access, occupancy, and crawlspace availability, where applicable, before travel.
- Where site conditions prevent complete capture, including weather, access, safety, or standing water in a crawlspace, the technician records each item that could not be observed and the reason. No item is estimated, assumed, or left blank.
- Where a crawlspace cannot be safely entered, the technician does not enter it. The engineer decides from the evidence collected whether a determination is possible or the job requires a confined-space specialist.

8. Field inspection

The technician completes every item of Appendix B for the property, in the order listed, and follows the photo procedure printed at the head of that checklist.

- Before starting, the technician makes a rough sketch of the foundation footprint, marks North, and photographs the sketch.
- The technician ticks the item being photographed, photographs the checklist, then takes the photographs for that item, and moves to the next item the same way.
- An item that does not apply to the property is marked with the reason it does not apply.
- The counts and measurements called for on the checklist are recorded as counts and measurements, not as descriptions.
- The relative elevation survey is performed at every job, and the benchmark location and reading method are recorded, because a differential reading is not meaningful without them.

9. Evidence standard and engineer review

Evidence package

Each job produces one evidence package in the firm's field application, consisting of the completed Appendix B checklist, the photographs, and the technician's notes.

- Every checklist item has at least one wide photograph placing the item on the structure and one close photograph showing its condition.
- Where a crack width, elevation differential, or gap governs, a ruler, crack comparator card, or the elevation survey reading is in frame or recorded alongside the photograph.
- The worst condition on each side of the foundation is photographed. A representative sample selected for appearance is not acceptable.
- Every crack, prior repair location, and foundation penetration is photographed individually.
- Photographs are captured directly in the firm's field application and carry device time, server time, the difference between them, and location. Photographs sent by text or messaging app, or taken on another device, are not accepted.
- An observation that cannot be made is recorded as not observed with the reason.
- A package that does not cover every checklist item, with the photographs and readings each item calls for, is rejected without review.

Engineer review

- The engineer reviews each evidence package and records one of the five determinations in Appendix C: pass, revise, repairs required, site revisit, or decline.
- The engineer's review record identifies the checklist items and photographs relied on for the determination.
- A package that does not allow the engineer to verify a required item is returned for revision or a site revisit. The customer is not charged for a re-inspection caused by an incomplete package.

10. Deficiencies and repairs

- On a repairs required determination, the engineer issues a written repair list identifying each item, what governs it, and what is required to close it.
- Repairs are performed by others. 254 Engineering Services does not perform, supervise, or recommend a contractor for the repair work.
- Corrections are verified on a site revisit to the same evidence standard, and the engineer closes each item individually.

- No certification letter is issued with an open repair item. There is no conditional certification.

11. Certification letter

- On a pass determination, the engineer issues a sealed letter addressed to the recipient recorded at intake, for the purpose recorded at intake.
- The letter identifies the property, the date of inspection, the foundation type, the method by which the evidence was collected and reviewed, and the observed condition.
- The letter states observed condition only. It does not forecast future movement, warrant against future movement, or represent that the foundation will not require future repair.
- The letter states that the inspection was visual and non-invasive, that it reflects condition on the date of inspection only, and that no representation is made about concealed or below-grade conditions.
- The letter is not reassigned to another recipient, another purpose, or another date. A new inspection is required.
- The engineer does not certify a foundation that was not inspected under this protocol by this firm.

12. Exclusions

The following are declined at this service tier or routed to another service line under its own engagement terms. The coordinator records the routing in the job file and informs the customer in writing.

- An open insurance claim on the foundation, where a certification would function as claim leverage. Declined, with a referral to an independent firm.
- Active or threatened litigation involving the foundation. Declined.
- A foundation under ongoing new construction, not yet complete. Declined — the firm does not offer inspection of foundations under construction at this time.
- A manufactured home foundation. Routed to the manufactured home foundation certification line.
- A request that is effectively for a geotechnical opinion, soil testing, or subsurface investigation beyond visual assessment. Declined, with a referral to a geotechnical engineer.
- Unsafe or inaccessible crawlspace that prevents minimum capture where the customer refuses to remediate access. Declined.
- A customer request for a predetermined conclusion. This is an automatic decline and the request is logged verbatim.

13. Records and retention

- Each job file contains: the intake record and customer uploads, the engineer's acceptance record, the dispatch record, the completed Appendix B checklist, all photographs, the technician's notes, the engineer's review record and determination, any repair list and its closure records, and the issued letter.
- Job files are retained for ten years from the date of the letter. For a job that produces no letter — a decline, or a file closed without a determination — the ten years runs from the date of the engineer's last recorded determination or closure.
- Ten years is the firm's floor, reviewed against TBPELS rules.
- The compliance file holds the engineer's license records, technician qualification and training records, and every issued version of this protocol.

14. Protocol maintenance

- The engineer reviews this protocol annually, and within thirty days of any change to the adopted code, to a governing industry guideline, or to the firm's field application.
- Revisions are issued as a new version number with the effective date and the engineer's approval recorded in the document control table.
- This document follows the firm's house protocol format. The portal's four parts are carried in Appendices A through C — Appendix A carries both the intake questions and the customer uploads. Appendix D is the reference list.

Appendix A. Intake questions and customer uploads

Asked and collected by the coordinator before the job is created. A job is not dispatched until every required item is present.

PART 1 — INTAKE QUESTIONS

Purpose & recipient

1. What is the letter for? (insurance binding/renewal / home sale / lender / permit / other — record exactly)

Answer: _____

2. Who will the letter be addressed to? (insurer name, buyer, lender, city)

Answer: _____

3. What deadline is the customer working against, if any?

Answer: _____

Property basics

4. Property address and county

Answer: _____

5. Property type (single family / duplex / townhome / small commercial) and number of stories

Answer: _____

6. Year built; foundation type (slab-on-grade / pier-and-beam / crawlspace); year of any prior foundation repair

Answer: _____

7. Post-tensioned or conventionally reinforced slab, if known

Answer: _____

Flags — a YES on any of these routes to engineer review BEFORE dispatch

8. Is there an OPEN insurance claim on this foundation?

Yes ☐ No ☐ If yes, describe: _____

9. Is there active or threatened litigation involving the foundation?

Yes ☐ No ☐ If yes, describe: _____

10. Any known active plumbing leak, slab leak, or recent leak history under the structure?

Yes ☐ No ☐ If yes, describe: _____

11. Has another engineer or inspector already issued an adverse report on this foundation?

Yes ☐ No ☐ If yes, describe: _____

12. Has the foundation been repaired before (piers, mudjacking, grouting)?

Yes ☐ No ☐ If yes, describe: _____

If yes — date, method, and company, if known: _____

13. Was a tree within 20 feet of the foundation removed in the last two years?

Yes ☐ No ☐ If yes, describe: _____

Access & safety (decides dispatch feasibility)

14. Is crawlspace access available and accessible, if applicable?

Answer: _____

15. Occupied or vacant? Accessible if occupied?

Answer: _____

16. Is the full foundation perimeter visible, or is it obstructed by decking, landscaping, or additions?

Answer: _____

Prior paperwork prompt

17. Do you have: prior foundation certs, repair invoices/warranty, permits, geotechnical/soils report, plumbing leak test report, elevation survey, photos of any problems?

Answer: _____

PART 2 — CUSTOMER UPLOADS (collected at intake, before dispatch)

Required

– Photo of the front of the property (confirms address/structure)

Required if it exists

– Any adverse report, prior engineer letter, insurer letter, or prior foundation repair warranty

Collect when applicable

– Foundation repair contract, invoice, or warranty for the most recent repair work (if repair claimed)

– Permits for foundation repair; HOA approval if applicable

– Geotechnical or soils report, if one exists

– Plumbing leak detection test report, if one exists

– Prior elevation survey or foundation inspection report

– Customer's own photos of problem areas (cracks, doors that don't close, etc.)

Appendix B. Technician on-site checklist

Completed on site by the technician. The checklist cannot be submitted incomplete.

PHOTOS

– Before starting, make a rough sketch of the foundation footprint and mark North. Take a photo of it.

– Tick the item you are about to photograph, take a photo of the checklist, then take the photos for that item.

– Move on to the next item the same way.

JOB INFORMATION

Job / Dispatch No.: _____

Property address (full, incl. county):

Technician name: _____

Date / arrival time / departure time: _____

Weather during visit: _____

Contact on site (name, role, phone): _____

☐ Photo of address marker (house number, street sign, or mailbox) [PHOTO]

CONTEXT (every job)

☐ All four elevations of the structure [PHOTO]

☐ Full perimeter foundation walk, continuous coverage [PHOTO]

Foundation type / shape / stories: _____

☐ Site grading overview — general lot slope relative to the structure [PHOTO]

☐ Each tree within 20 feet of the foundation — species and trunk diameter if known [PHOTO]

Technician notes / readings:

EXTERIOR FOUNDATION CONDITION

☐ Every visible crack: wide + close with ruler or crack comparator card in frame; note orientation (vertical/horizontal/diagonal/stair-step), width, length, and location [PHOTO]

☐ Stair-step cracking in brick veneer or mortar joints — count per elevation, worst close-ups [PHOTO]

☐ Separation at expansion joints, control joints, or where foundation meets siding/veneer [PHOTO]

☐ Spalling, exposed rebar, or exposed post-tension cable/anchor [PHOTO]

☐ Efflorescence or moisture staining on the foundation face [PHOTO]

☐ Foundation vent condition and screening (pier-and-beam / crawlspace) [PHOTO]

☐ Downspout discharge points and distance from foundation [PHOTO]

☐ Standing water or ponding evidence adjacent to the foundation [PHOTO]

☐ Irrigation heads or hose bibs discharging directly at the foundation [PHOTO]

☐ Drainage system if present — French drain, area drain, or swale, and its discharge point [PHOTO]

Counts — cracks over 1/16 in. ____ stair-step cracks ____ drainage deficiencies ____: _____

Technician notes / readings:

RELATIVE ELEVATION SURVEY

☐ Survey equipment and benchmark location, photographed in place [PHOTO]

Equipment used (laser level / water level / Zipline / other): _____

Benchmark location (e.g., interior door threshold nearest lowest point):

☐ Grid reading points across each room and along the perimeter, marked on the sketch, spaced no more than 20 ft apart [PHOTO]

☐ Photo of the reading or device display at each grid point, or the summary printout [PHOTO]

Overall differential, full length / width / diagonal (report whichever governs):

Differential (in.): _____ Span measured (ft): _____ Direction (length / width / diagonal):

Largest differential between any two adjacent grid points:

Differential (in.): _____ Spacing between points (ft, 20 ft or the foundation dimension, whichever is less): _____

Technician notes / readings:

INTERIOR FINISH SCAN

☐ Interior wall and ceiling cracks, especially at corners and above door/window headers — wide + close with ruler [PHOTO]

☐ Floor slope or tilt — level or ball-roll test at each room tested, wide shot of test [PHOTO]

☐ Every interior door tested for function — binding, gaps, fails to latch — photo of each affected door and frame [PHOTO]

☐ Every window tested for function — binding or fails to latch — photo of each affected window [PHOTO]

☐ Flooring separation or tile cracking, especially near walls — wide + close [PHOTO]

☐ Baseboard or trim gaps at floor or wall junctions [PHOTO]

Doors tested ____ doors with function issues ____ windows tested ____ windows with function issues ____

Technician notes / readings:

CRAWLSPACE / SUBFLOOR VERIFICATION (pier-and-beam only)

- ☐ Overview of the crawlspace from the access point [PHOTO]
- ☐ Every pier — tilting, settlement, cracking, or loss of contact with the beam — wide + close of any affected pier [PHOTO]
- ☐ Beam and girder condition — rot, sagging, cracking — worst examples with ruler [PHOTO]
- ☐ Joist condition and spacing — deterioration or damage [PHOTO]
- ☐ Standing water, active moisture, or saturated soil in the crawlspace [PHOTO]
- ☐ Vapor barrier presence and condition, if installed [PHOTO]
- ☐ Ventilation — vent count and any blocked vents [PHOTO]
- ☐ Wood-destroying insect evidence — mud tubes, frass, damaged wood [PHOTO]
- ☐ Evidence of prior pier replacement, shimming, or repair — tags, plates, or contractor markings [PHOTO]

Pier count observed ____ piers with a deficiency ____: _____

Technician notes / readings:

FOUNDATION AGE AND PRIOR REPAIR VERIFICATION

- ☐ Any foundation repair company signage, tags, or plates found on site [PHOTO]

Owner-reported foundation age / prior repair year (verbatim):

- ☐ Any visible new cracking or movement near a previously repaired zone [PHOTO]

Technician notes / readings:

CLOSE-OUT

- ☐ Full perimeter walk — anything unusual [PHOTO]
- ☐ Exception menu cleared for every non-applicable item

Total photos: _____

Technician signature / date: _____

Technician notes / readings:

Appendix C. Engineer decision rules

Applied by the engineer to each evidence package. One determination is recorded per review.

PASS

- Package complete: every checklist item present or properly excepted
- No active moisture or plumbing leak evidence at the foundation or in the crawlspace
- Cracks are hairline only on a foundation under 10 years old, or no wider than 1/8 in. on a foundation 10 years or older, with no active displacement
- Foundation slope no more than 1% in any direction — about 1 inch of rise or fall over every 8 feet, checked across the length, width, and corner-to-corner — and no single stretch between two survey points shows more than about 1 inch of difference over a 30-foot span
- Doors and windows function normally, or show only minor issues consistent with ordinary seasonal movement
- Drainage and grading adequate, or noted as a maintenance item only

REVISE

- Any required photo missing, blurry, or missing its ruler/comparator-card pair
- Elevation survey grid too sparse to check the slope, or missing the benchmark record
- Worst-condition coverage doubtful (all photos look selectively favorable)
- Exception used where the condition plainly applied
- Counts or measurements inconsistent with photos

REPAIRS REQUIRED (certification withheld, repair list issued)

- Active moisture or plumbing leak evidence under the structure
- A crack wider than hairline on a foundation under 10 years old, or wider than 1/8 in. on a foundation 10 years or older, at a structural location, or any crack showing active displacement — this includes a separation gap at an addition or adjoining structure, measured the same way
- Foundation slope over 1% in any direction — more than about 1 inch of rise or fall over 8 feet, across the length, width, or corner-to-corner
- Any single stretch between two survey points with more than about 1 inch of difference over a 30-foot span
- Pier or beam failure, or confirmed loss of pier-to-beam contact (pier-and-beam)
- Stair-step cracking with visible rotation or displacement of veneer
- Certification proceeds only after repairs are verified on revisit

SITE REVISIT (technician returns with an engineer-specified capture list)

- Elevation survey incomplete or grid too sparse, return to complete the grid
- Crack present without a comparator-card measurement, return with the tool and re-photograph
- Suspected active movement without monitoring data.
- Conflicting evidence (owner story vs. photos vs. uploads), return for the specific items that settle it: repair records, permit, soils report
- Prior adverse report exists, return and photograph every condition the report cites at the same locations before contradicting it
- Repairs required, return after repairs to verify each item on the repair list

DECLINE

- Open insurance claim where the certification would function as claim leverage — decline, with a referral to an independent firm
- Active or threatened litigation — decline
- Foundation under ongoing new construction, not yet complete — decline; the firm does not offer this service at this time
- Manufactured home foundation — route to manufactured home foundation certification line
- Request is effectively for a geotechnical opinion or subsurface investigation — decline, with a referral to a geotechnical engineer
- Unsafe or inaccessible crawlspace that prevents minimum capture and customer refuses to remediate access
- Customer requests a predetermined conclusion — automatic decline, log verbatim request

Appendix D. References

Item	Location
2024 International Residential Code, Chapter 4, Foundations	As adopted by the authority having jurisdiction
2024 International Building Code, Chapter 18, Soils and Foundations	As adopted by the authority having jurisdiction
Texas Engineering Practice Act, Texas Occupations Code Chapter 1001	statutes.capitol.texas.gov
TBPELS rules, 22 TAC Chapter 137	pels.texas.gov
FPA-SC-07-0, Foundation Maintenance and Inspection Guide for Residential and Other Low-Rise Buildings, Foundation Performance Association	foundationperformance.org/projects/FPA-SC-07-0.pdf
FPA-SC-12-0, Guidelines for Evaluating Foundation Performance by Monitoring, Foundation Performance Association	foundationperformance.org/projects/FPA-SC-12-0.pdf
FPA-SC-13-0, Foundation Movement Evaluation Guidelines, Foundation Performance Association	foundationperformance.org/projects/FPA-SC-13-0.pdf
Post-Tensioning Institute — field guidance, where a post-tensioned slab is identified	post-tensioning.org

254-RC-001 Roof Certification Protocol, v1.1, issued September 18, 2026	Firm document
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Approval

Reviewed and approved by the Engineer of Record. This protocol is in force from the date below.

Aman Dhakal, P.E.

Engineer of Record, 254 Engineering Services

Signature: _____ Date: 2026-09-29

AD

CERTIFICATE *of* SIGNATURE

REF. NUMBER
GDBJF-FU8RW-95HLZ-WHZBQ

DOCUMENT COMPLETED BY ALL PARTIES ON
29 SEP 2026 12:38:27
UTC

SIGNER

AMAN DHAKAL

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TIMESTAMP

SENT
29 SEP 2026 02:49:51

VIEWED
29 SEP 2026 12:38:01

SIGNED
29 SEP 2026 12:38:27

SIGNATURE



IP ADDRESS
104.28.50.136

LOCATION
FORT WORTH, UNITED STATES

RECIPIENT VERIFICATION

EMAIL VERIFIED
29 SEP 2026 12:38:01

